

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
= 1	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Avg power density $l/\lambda < 1/50$; far field	
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$ (W/m ²)	
<i>Equivalent:</i> $S = S_{\text{max}} \sin^2 \theta$ with $S_{\text{max}} = \eta_0 k^2 I_0^2 l^2 / (32\pi^2 R^2)$ at $\theta = \pi/2$.	
<i>l</i> : rod length (m) <i>R</i> : obs. distance (m) <i>k</i> = $2\pi/\lambda$ (rad/m)	<i>I</i> ₀ : peak current (A) <i>θ</i> : angle from dipole axis (rad) <i>η</i> ₀ = 120π ≈ 377 Ω
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density any l; far field	
$S(R, \theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$ (W/m ²)	
<i>At θ = 0, π: bracket is 0/0. L'Hôpital ⇒ S = 0 (no axial radiation). TI-36X alt: evaluate bracket at θ = 0.001 rad → tiny → 0.</i>	
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$ $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$						
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	
0	0	0	1	0	1	
30	π/6	1/2	√3/2	1/4	3/4	
45	π/4	√2/2	√2/2	1/2	1/2	
60	π/3	√3/2	1/2	3/4	1/4	
90	π/2	1	0	1	0	
180	π	0	−1	0	1	

Antenna formulas usually take *θ* in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$	
Normalize $F_a^{\text{norm}} = F_a/N^2$	
HPBW (broadside, uniform) use Nd, not $(N-1)d$ $\beta \approx 0.886 \frac{\lambda}{Nd}$ (rad) = 50.8° · $\frac{\lambda}{Nd}$	
Electronic steering (scan to θ_0) replace γ with $\gamma' = kd(\cos \theta - \cos \theta_0)$ $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elt}}$); $\theta_0 = \frac{\pi}{2}$: broadside, 0: end-fire	
Frequency scan feed-line incr. $\Delta \ell = n_0 \lambda_0$ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$	
<i>Grating lobes appear if $d > \lambda$. Avoid.</i>	

DIPOLE

<i>l</i> — antenna length (m) $\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s <i>I</i> ₀ — peak current (A) <i>k</i> = $2\pi/\lambda$ (rad/m) <i>R</i> — obs. dist. (m); <i>θ</i> from axis <i>η</i> ₀ = 120π ≈ 377 Ω <i>S</i> (<i>R, θ</i>) — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless) <i>a</i> — coef. on θ^2 in Gaussian <i>β</i> — HPBW (rad or °) <i>Ω</i> _p — pattern solid angle (sr) <i>D</i> = $4\pi/\Omega_p$ — directivity <i>ξ</i> radiation eff. (unitless); <i>G</i> = <i>ξD</i> $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIS / dB

<i>P</i> _t — power transmitted (W) <i>P</i> _r — power received (W) <i>G</i> _t — transmit gain (linear) <i>G</i> _r — receive gain (linear) <i>S</i> _r = <i>G</i> _t <i>P</i> _t /(4π <i>R</i> ²) (W/m ²) <i>P</i> _r = <i>G</i> _t <i>G</i> _r (<i>λ</i> /4π <i>R</i>) ² <i>P</i> _t <i>G</i> = 10 ^{<i>G</i> dB/10} 3 dB = ×2, 10 dB = ×10 <i>P</i> _N = <i>k</i> <i>B T</i> _{sys} <i>B</i> (W)
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APERTURE

<i>A</i> _p — physical area (m ²) <i>A</i> _e = <i>λ</i> ² <i>D</i> /(4π) (m ²) <i>A</i> _e ≈ <i>A</i> _p for aperture <i>D</i> ≈ 4π <i>A</i> _p / <i>λ</i> ² (unitless) <i>β</i> ≈ 0.88 <i>λ</i> / <i>ℓ</i> (rect/sq.) <i>β</i> ≈ 1.02 <i>λ</i> / <i>d</i> (circular) 2 <i>A</i> _p ⇒ 2 <i>D</i> , <i>β</i> /√2
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ARRAY

<i>N</i> — number of elements <i>d</i> — spacing (m) <i>γ</i> = <i>k d cos θ</i> <i>F</i> _a = $\sin^2(N\gamma/2) / \sin^2(\gamma/2)$ <i>F</i> _a ^{max} = <i>N</i> ² at <i>θ</i> = π/2 <i>F</i> _a ^{norm} = <i>F</i> _a / <i>N</i> ² <i>β</i> ≈ 0.886 <i>λ</i> /(<i>Nd</i>) (broadside)

BEAM PARAMETERS — *β, Ω_p, D*

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless) Beamwidth <i>β</i> at threshold <i>X</i> Set $F(\theta_X) = X$, solve for <i>θ_X</i> ; <i>β</i> = 2 <i>θ_X</i> (symmetric)		
Threshold	<i>X</i> = 10 dB/10	<i>θ_X</i>
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any <i>F</i> (<i>θ_X</i>)	any <i>X</i>	<i>F</i> (<i>θ_X</i>) = <i>X</i>
Pattern solid angle <i>Ω_p</i> recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr) <i>No φ dep.: Ω_p = 2π ∫₀^π F(θ) sin θ dθ.</i> <i>θ is the integration variable — DON'T plug in a number for θ (e.g. θ_{1/2}). Integrate over θ ∈ [0, π].</i>		
Pattern <i>F</i> (<i>θ</i>)	<i>Ω_p</i> (sr)	
Isotropic (<i>F</i> = 1)	4π ≈ 12.57	
Hertzian <i>sin² θ</i>	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss <i>e^{−aθ²}</i>	π/ <i>a</i> = π <i>θ</i> ₂ ² /ln 2	
2-axis HPBW (aperture)	<i>β_{xz}</i> · <i>β_{yz}</i>	
Other (use integral)	numerical	

TI-36X: [2nd] [d/dx] for $\int \square$, RAD mode, multiply by 2π.

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below		
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{dB} = 10 \log_{10} D$	
WORD TRAP: “direction” = $\theta_{\max}$ (rad); “directivity” = D (unitless).		
Gain (if asked) $\xi =$ radiation efficiency (unitless), $0 < \xi \leq 1$	$G = \xi D$ (unitless) $G_{dB} = 10 \log_{10} G$ (dB)	
Lossless / ideal antenna $\Rightarrow G = D$.		
$\xi = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$; $R_{\text{rad}} = 2P_{\text{rad}}/I_0^2$ (Ω); $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$.		

COMMON DIPOLE PARAMETERS

Lossless (<i>ξ</i> = 1): <i>G</i> = <i>D</i> .				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< <i>λ</i> /50	1.5	1.76	80π ² (<i>l</i> / <i>λ</i>) ²
Half-wave	<i>λ</i> /2	1.64	2.15	73.2
Monopole	<i>λ</i> /4 (gnd)	1.64	2.15	36.6
Full-wave	<i>λ</i>	2.41	3.82	199
Half-wave: <i>P</i> _{rad} = 36.6 <i>I</i> ₀ ² W. Monopole (<i>λ</i> /4 above ground): same patt. as half-wave but <i>P</i> _{rad} , <i>R</i> _{rad} halved.				

ANTENNA AREAS (*A_e, A_p*)

Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter <i>d</i>; <i>l</i> see table above <i>A</i> _p = <i>l d</i> (m ² , any dipole)	
Inverse: <i>D</i> from <i>A_e</i> $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire:</i> <i>A_e</i> ≫ <i>A_p</i> (e.g., Hertzian <i>A_e</i> = 3 <i>λ</i> ² /8π ≈ 0.119 <i>λ</i> ²).	

FRIS TRANSMISSION FORMULA

<i>For each antenna's G: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (G = D, lossless). Else (dB/linear value given) → use as given (convert dB → linear).</i>	
Pwr density at RX, then RX pwr <i>G_t, G_r</i> linear gains; <i>λ</i> = <i>c</i> / <i>f</i>	$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } \left[P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \right] \text{ (W)}$
<i>Equivalent:</i> <i>P_r</i> = <i>S_r</i> <i>A_{e,r}</i> .	
Solve for <i>R</i> range	$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$
Equivalent form using <i>A_e</i> recv. area form $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$	
Free-space path loss: <i>L_{fs}</i> = (4π <i>R</i> / <i>λ</i>) ² , so <i>P_r</i> = <i>G_t</i> <i>G_r</i> <i>P_t</i> / <i>L_{fs}</i> .	
Off-axis (patterns not aligned): multiply by <i>F_t</i> (<i>θ_t</i> , <i>φ_t</i>) <i>F_r</i> (<i>θ_r</i> , <i>φ_r</i>), both = 1 when aligned.	
Recipe 1. <i>λ</i> = <i>c</i> / <i>f</i> . 2. Convert any dB gain → linear: <i>G</i> = 10 ^{<i>G</i> dB/10} .	
3. Get <i>G_t, G_r</i> by antenna type: dipole→COMMON DIPOLE PARAMS; aperture→APERTURE box; arbitrary <i>F</i> →BEAM PARAMS. 4. Plug into Friis.	
<i>Use linear G, not dB. Convert FIRST.</i>	

dB ↔ LINEAR

<i>Works for any power-like quantity (G, D, P_r/P_t, SNR, ...):</i>	
<i>X</i> _{dB} = 10 log ₁₀ <i>X</i> _{lin} <i>X</i> _{lin} = 10 ^{<i>X</i>_{dB}/10}	
<i>Memorize:</i> 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.	
<i>For E-field / amplitude use 20 log not 10 log (power ∝ E²).</i>	

NOISE & SNR

Noise power thermal noise into matched receiver <i>P_N</i> = <i>k</i> <i>B T</i> _{sys} <i>B</i> (W)	
<i>k</i> _B = 1.38×10 ^{−23} J/K; <i>T</i> _{sys} system noise temp (K); <i>B</i> receiver bandwidth (Hz).	
Signal-to-noise ratio	
$\text{SNR} = \frac{P_{\text{rec}}}{P_{\text{N}}} \text{ (unitless)} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_{\text{N}}}$	

APERTURE ANTENNAS (horn, dish)

<i>A_p</i> physical aperture area (m ²); <i>A_e</i> effective area (m ² , = <i>λ</i> ² <i>D</i> /4π); <i>β_{xz}, β_{yz}</i> HPBW's in two planes (rad); <i>ℓ, d</i> aperture side / diameter (m).			
<i>Far-field req.: R</i> ≥ 2 <i>d</i> ² / <i>λ</i> (<i>d</i> = largest aperture dim.).			
Pattern (rect., uniform illum.) peak at θ = 0 (broadside) $F(\theta) = \text{sinc}^2 \left(\frac{\pi \ell_x \sin \theta}{\lambda} \right)$; $S_0 = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}$			
Directivity (any shape, uniform illum.) $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ <i>A_e</i> ≈ <i>A_p</i>			
Directivity from beamwidths any aperture $D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$			
HPBW and <i>D</i> by aperture shape uniform illum.; <i>β</i> in rad			
Shape	<i>A_p</i>	HPBW (rad)	<i>D</i> (unitless)
Rect. <i>ℓ_x</i> × <i>ℓ_y</i>	<i>β_x</i> ≈ 0.88 <i>λ</i> / <i>ℓ_x</i> <i>β_y</i> ≈ 0.88 <i>λ</i> / <i>ℓ_y</i>	<i>D</i> ≈ 4π/(<i>β_xβ_y</i>) = 4π <i>ℓ_xℓ_y</i> / <i>λ</i> ²	
Square <i>ℓ</i>	<i>β</i> ≈ 0.88 <i>λ</i> / <i>ℓ</i>	<i>D</i> ≈ 4π/ <i>β</i> ² = 4π <i>ℓ</i> ² / <i>λ</i> ²	
Circular (diam. <i>d</i>)	<i>π d</i> ² /4	<i>β</i> ≈ 1.02 <i>λ</i> / <i>d</i>	<i>D</i> ≈ 4π/ <i>β</i> ² = <i>π</i> ² <i>d</i> ² / <i>λ</i> ²
Scaling (any ratio) <i>D</i> ∝ <i>A_p</i> / <i>λ</i> ² ; <i>β</i> ∝ <i>λ</i> /√ <i>A_p</i>			
$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$			
$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$			
<i>If a parameter is unchanged, its ratio = 1. Examples: double area → D×2, β×1/√2; double freq → D×4, β×1/2.</i>			